
Lecture 1: Introduction to Machine Learning

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Artificial Intelligence

- Artificial Intelligence
 - The science of making machine that think like people and act like people
 - ability of a computer or machine to mimic or imitate human intelligent behavior and perform human-like tasks
- Machine Learning is a sub set of AI



Machine Learning

- Machine Learning is the science (and art) of programming computers so they can *learn from data*.

- More general definition:

Machine Learning is the field of study that gives computers the ability to learn without being explicitly programmed.

—Arthur Samuel, 1959

- Engineering-oriented definition:

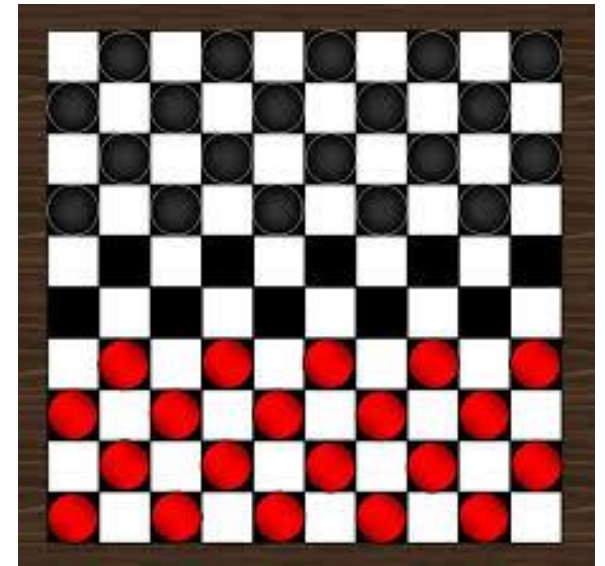
A computer program is said to learn from **experience E** with respect to some **task T** and some **performance measure P** , if its performance on T , as measured by P , improves with experience E .

—Tom Mitchell, 1997



Example

- Checkers learning problem
 - Task T : playing checkers
 - Performance measure, P : % of games won against opponents
 - Training Experience, E : Playing practice games against itself



Example

- Handwriting Recognition Problem
 - Task T : Recognizing and classifying handwritten words/characters within images
 - Performance measure P : % of words/characters correctly classified
 - Training Experience, E : Predicting the word/character from a collection of handwritten words/characters with given classifications
- Data Set-MNIST Dataset (Yann Lecunn et al.)



Example

- Classifying an incoming email as either a spam or not spam
 - Task T: Correctly classify an email as either spam or not spam
 - Performance measure, P: % of emails correctly classified
 - Training Experience, E: A database of emails with proper labeling as either spam or not.



Example

- Robot driving learning problem
 - Task T : driving on public highways using vision sensors
 - Performance measure, P : Average distance travelled before an error (as judged by a human observer)
 - Training Experience, E : A sequence of images and steering commands recorded while observing a human driver.



Traditional Programming

- Traditional Programming: writing a set of **RULES** to find **ANSWERS** from **DATA**

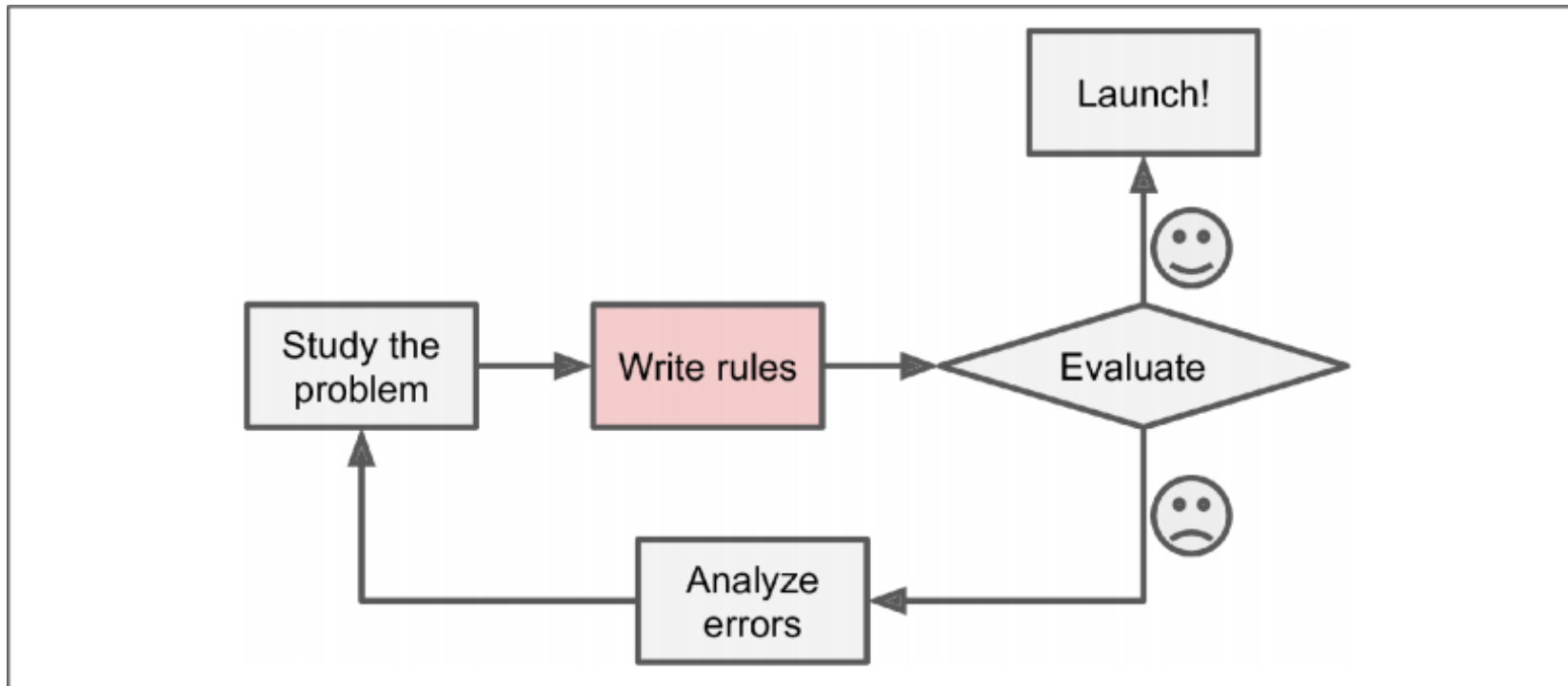


Figure 1-1. The traditional approach

The ML Approach

- **Machine Learning:** Use **DATA** and **ANSWERS** to learn the underlying set of **RULES**

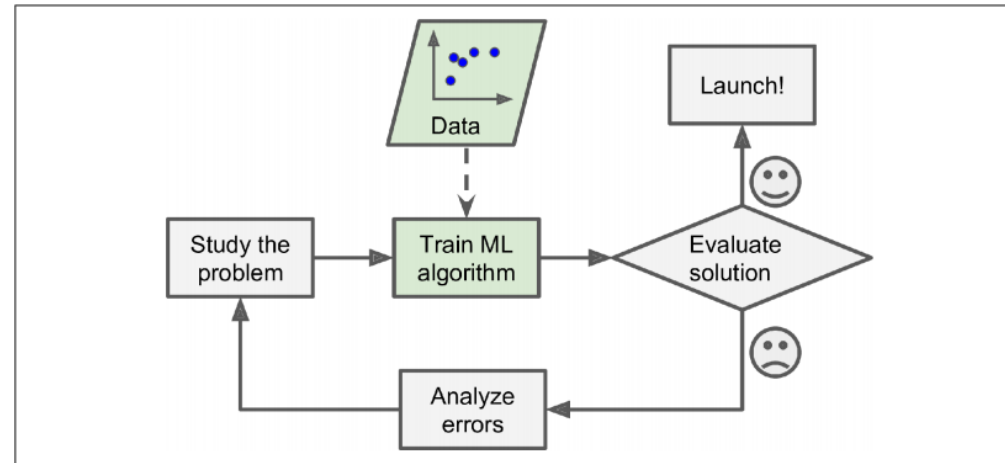


Figure 1-2. The Machine Learning approach

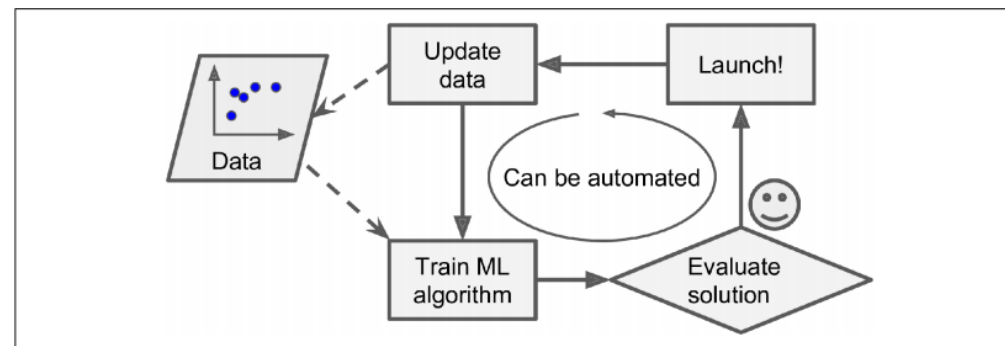


Figure 1-3. Automatically adapting to change

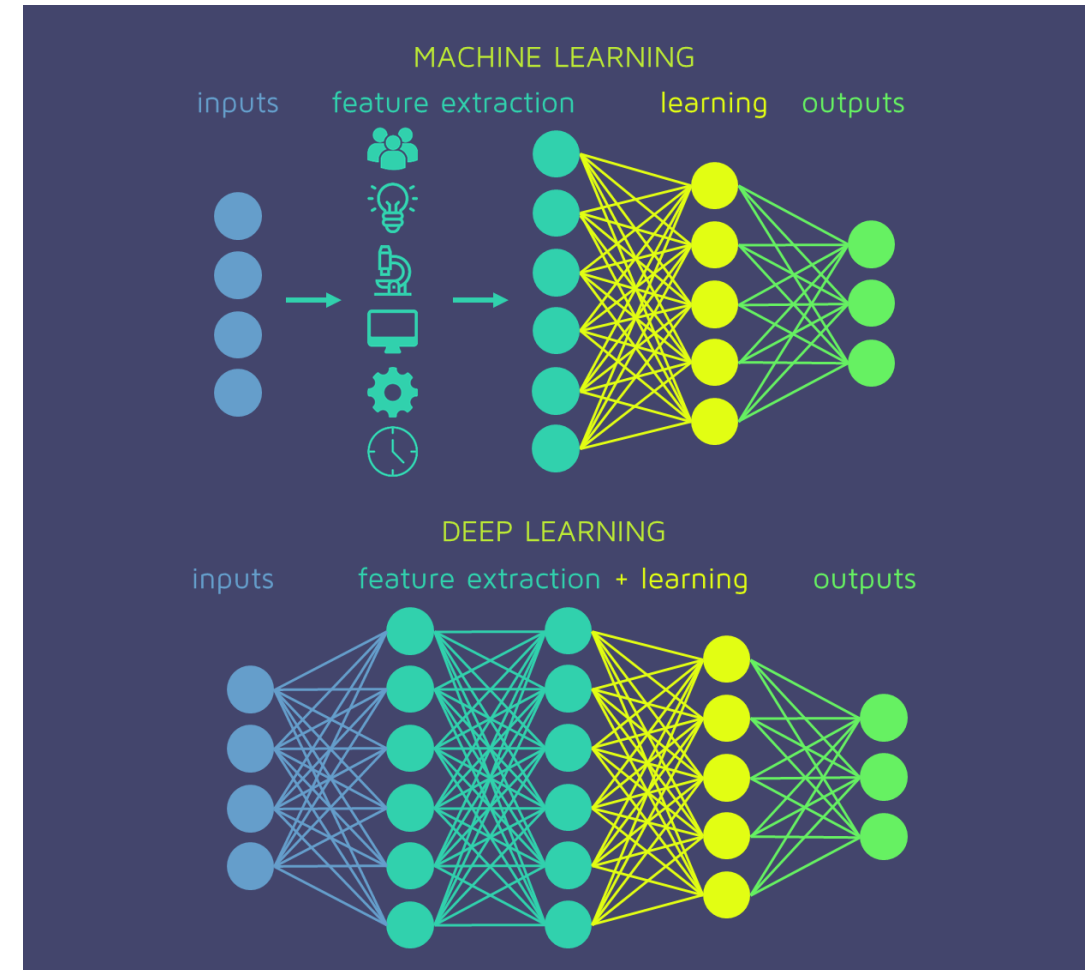
Benefits of ML using

- Problems that require a lot of **fine-tuning** or **long list of rules**
- **Changing environments** – ML systems can ADAPT
- Getting insights from **large amounts of data**
- **Complex** problems that yields no good solution with traditional approach



Deep Learning

- Subset of ML - loosely mimics structure/function of human brain
- Unlike traditional ML, does not require manual feature extraction
- Keeps getting better with more data (typically)



Types of Machine Learning Systems

- There are so many different types of Machine Learning systems that it is useful to classify them in broad categories, based on the following criteria:

1. Does it require human supervision?

- Supervised Learning
- Semisupervised Learning
- Unsupervised Learning
- Reinforcement Learning



Types of Machine Learning Systems

2. Can it learn incrementally on the fly?

- Online Learning
- Batch Learning

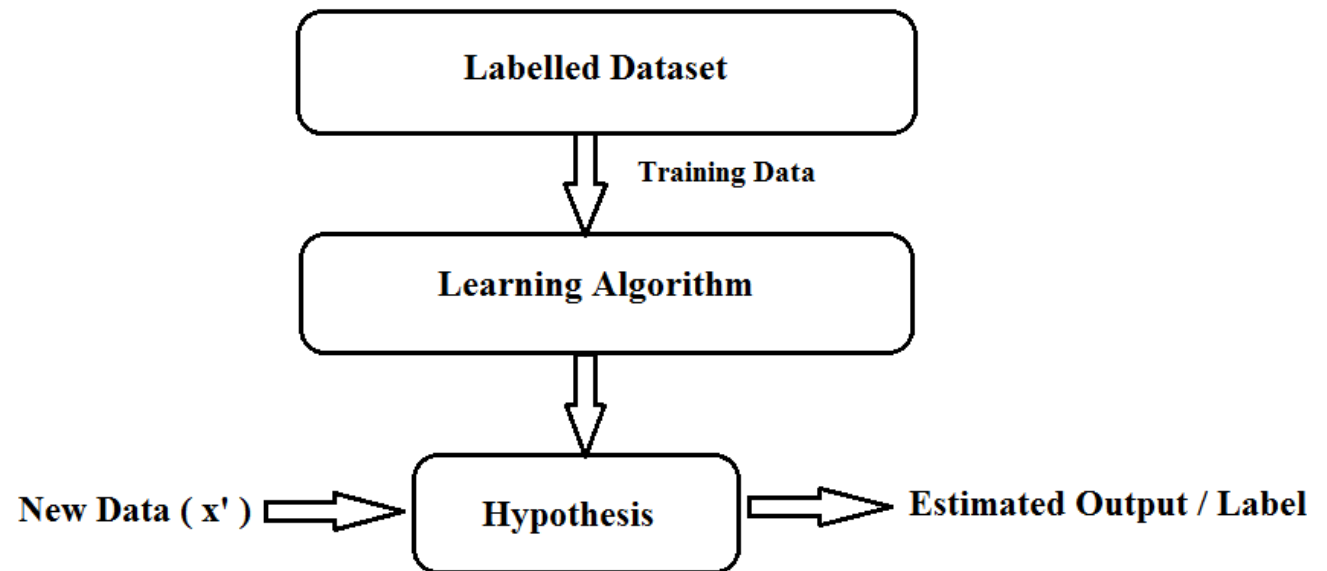
3. Does the system build a predictive model?

- Model-based Learning
- Instance-based Learning



Supervised Learning

- ML algorithm uses the training dataset to create **model/assumption/hypothesis** which can be used to predict (unseen/new data)
- Training data fed to algorithm includes the desired answers/solutions (**labels**)
- ML algorithm iteratively makes predictions on the training data and is corrected by the output/supervision of teachers



Supervised Learning

- Example algorithms:
 - Linear Regression
 - Logistic Regression
 - SVM
 - Decision Tree
 - Neural Network



Supervised Learning

Data:

- 1. Qualitative: Non-numeric/Categorical (hair color);
- 2. Quantitative (numeric)
 - Quantitative (numeric): Discrete (certain specific value); Continuous (any value)



Supervised Learning

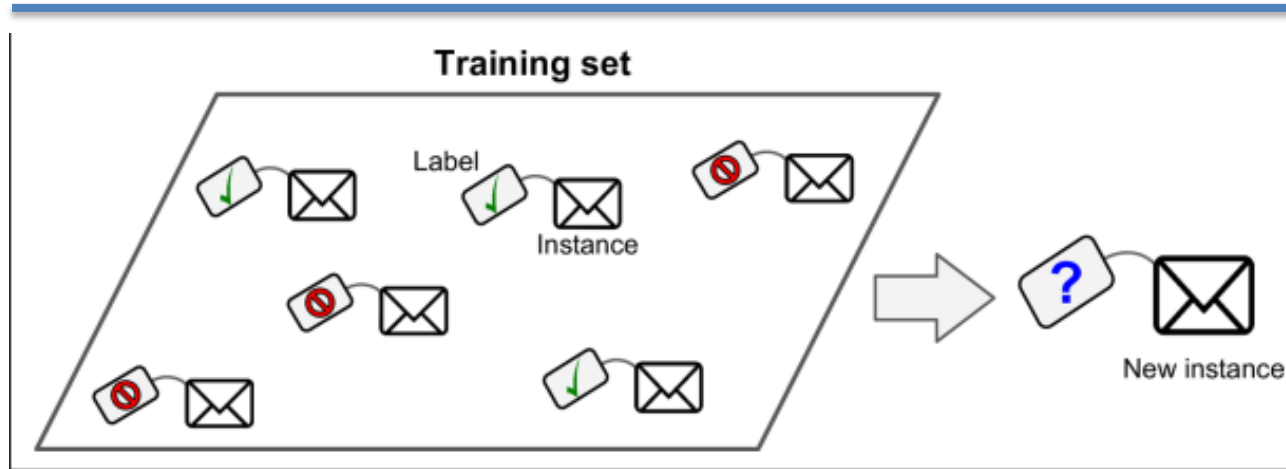
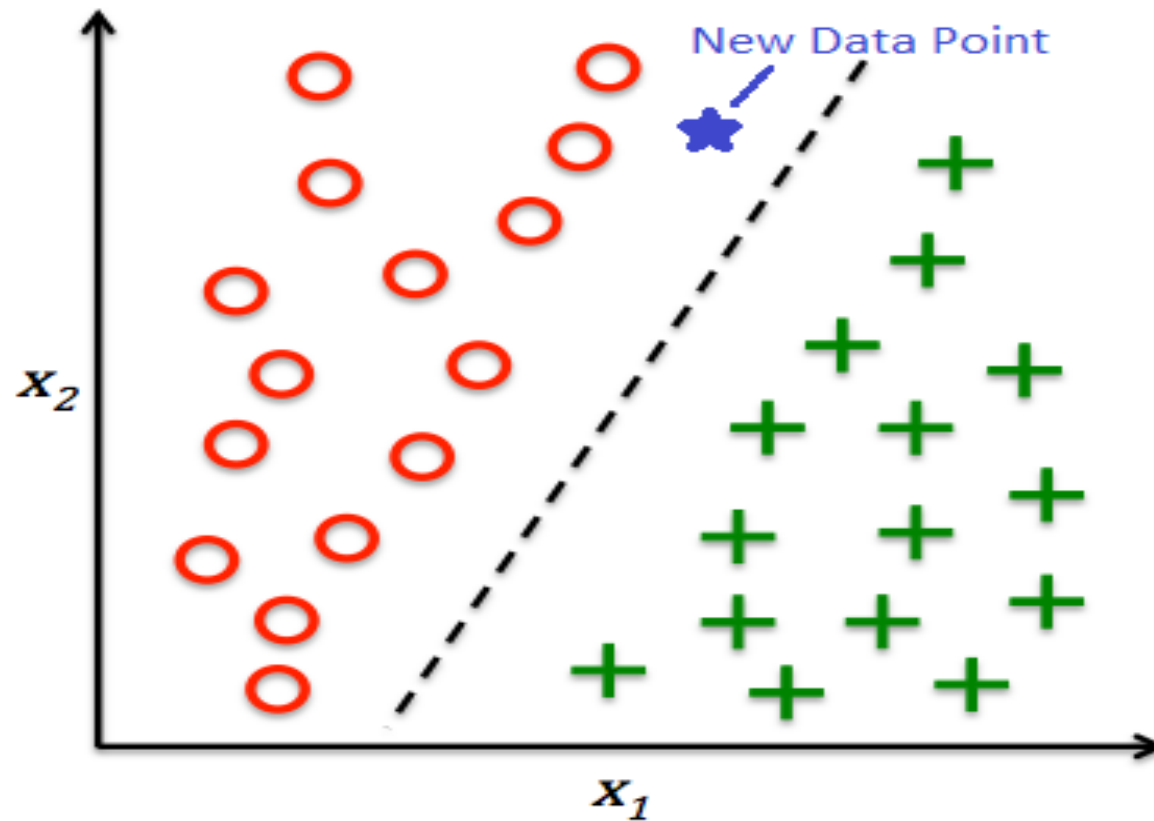


Figure 1-5. A labeled training set for spam classification (an example of supervised learning)



Figure 1-6. A regression problem: predict a value, given an input feature (there are usually multiple input features, and sometimes multiple output values)

Supervised Learning



Unsupervised Learning

- Unlabeled dataset. System learns without direct human supervision
- Unsupervised learning algorithms take a set of data that contains only inputs, and find structure/patterns in the data, like grouping or clustering of data points.
- The algorithms therefore learn from test data that has not been labeled, classified or categorized
- Instead of responding to feedback, unsupervised learning algorithms identify commonalities in the data and react based on the presence or absence of such commonalities in each new piece of data.
- Useful for business intelligence



Unsupervised Learning

- Example algorithms:
 - K-means
 - PCA (Principal Component Analysis)
 - SVD (Singular Value Decomposition)
 - ICA (Independent Component Analysis)
- Widely used in:
 - Clustering
 - Anomaly detection

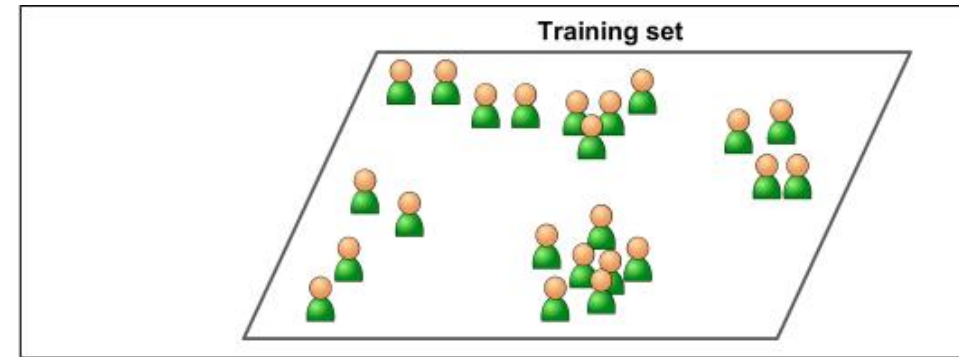


Figure 1-7. An unlabeled training set for unsupervised learning

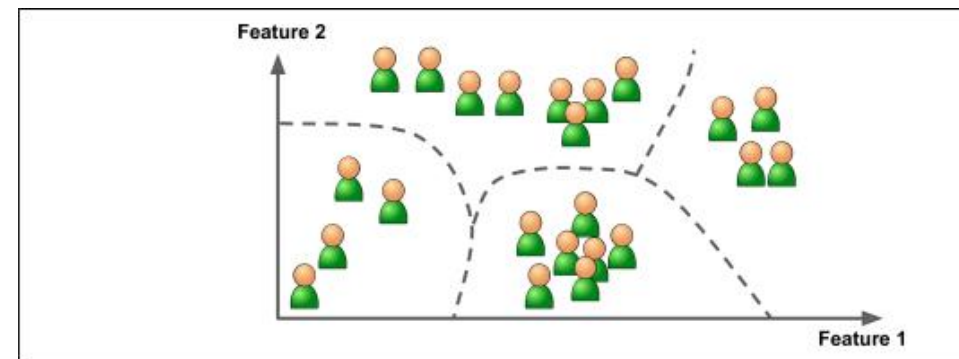


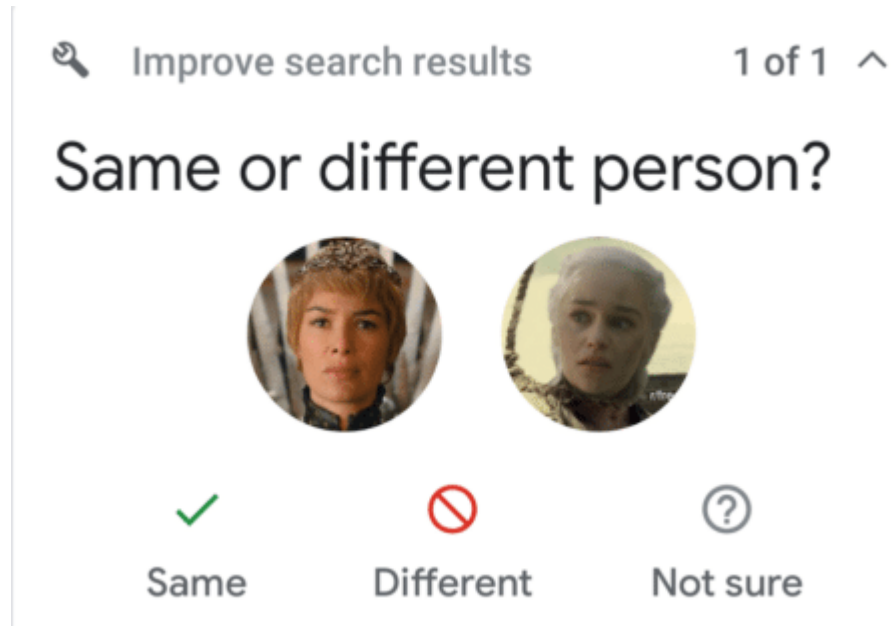
Figure 1-8. Clustering

Semi-Supervised learning

- In Semi-Supervised learning, the training data contains both **labeled and unlabeled data**
 - We have a large dataset. Manual labeling of the entire dataset is laborious and expensive
 - One solution could be to label a sample of the dataset and train the labeled portion to create a model. This would however mean that we are not fully utilizing the larger dataset we have and thus the model that we create may be less robust
 - A potential solution: (1) Label a sample of the large dataset, (2) train a model using this labeled portion, (3) use the model to predict the unlabeled portion (**pseudo-labeling**), (4) train using the entire dataset
 - e.g., Google Photos will cluster similar faces, and ask the user if they are the same person

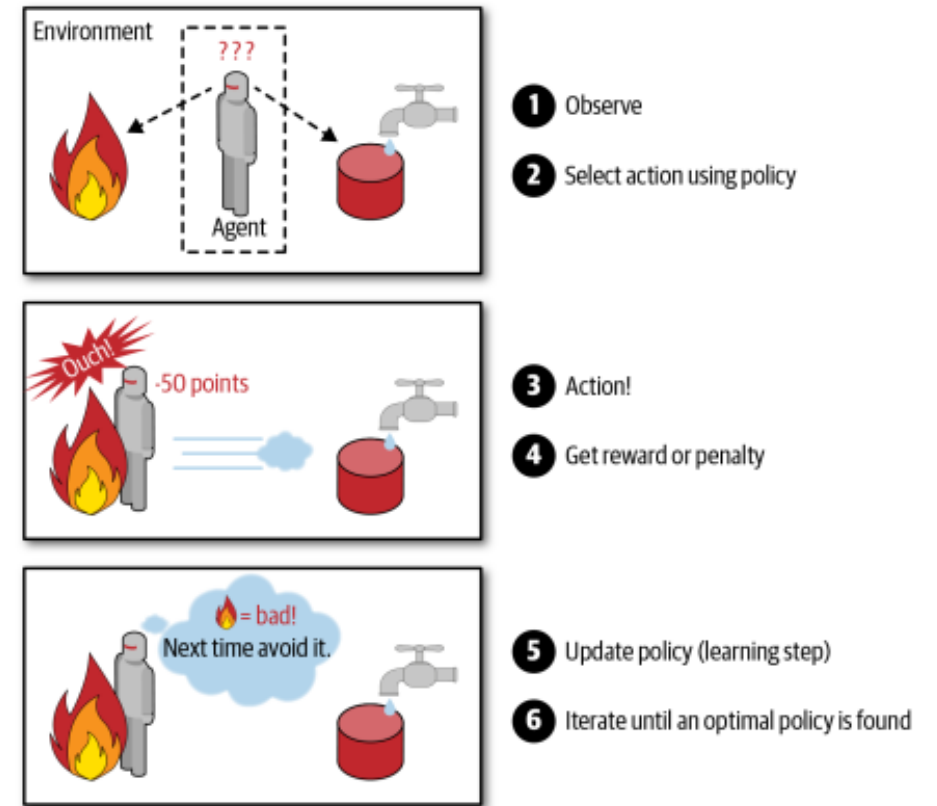


Semi-Supervised learning



Reinforcement Learning

- The learning system (agent) can:
 - Observe the environment
 - Select and perform an action based on environment
 - Get rewards/penalties as a result
 - Based on the reward agent will change its state
 - **Agent aim: Maximize reward**
- Learns what the best policy should be
 - Policy defines what actions should be chosen in a certain situation
- Very effective in controlled environments (such as a game of chess)
 - With the progress in deep learning, increasingly used in more complex tasks (such as driving the mars rover)



Reinforcement Learning

- Reinforcement learning differs from supervised learning in a way that in supervised learning the training data has the answer key with it so the model is trained with the correct answer itself
- whereas in reinforcement learning, there is no answer but the reinforcement agent decides what to do to perform the given task.
- In the absence of a training dataset, it is bound to learn from its experience

